

There are many definitions of risk, but the definition I've found that most closely aligns with investing and risk management is that risk is the potential for deviation from an expected result, whether the deviation is a positive or negative result. Risk management, by extension, is the analytical process that measures the potential for deviation in order to better enable decision-making on what risk to pursue, mitigate or avoid. Risk and return go hand-in-hand and every investment decision carries an element of risk, which is why a practical understanding of financial theory can assist investment decision-making and risk management.

Capital Asset Pricing Model (CAPM)

Much has been written about CAPM over the years, and like all models it is built from limiting assumptions. To start with, financial theory itself assumes securities markets are competitive, efficient, and comprised of rational investors who are seeking to maximize their return. From these two assumptions, CAPM adds additional assumptions including: no transaction costs, no taxes, and limitations on short selling.

Understanding these assumptions allows investors and risk managers to put in perspective how CAPM works, how it can be used, and, even with limiting assumptions, CAPM provides a model of the financial markets that measures risk and expected return, and underpinning CAPM is that risky assets can be combined in a portfolio such that the portfolio is less risky than any individual security.

Since most securities tend to have some degree of correlation it is not possible to completely eliminate risk through diversification. Accordingly, total risk of a security is comprised of systematic risk and unsystematic risk. Systematic risk cannot be diversified away because it is related to the overall market itself, but unsystematic risk is specific to the company and it can be reduced by diversification. Studies have shown that unsystematic risk can be significantly reduced in portfolios comprised of at least 30 randomly selected securities. It does need to be pointed out, however, that this assumes the selected securities are not in closely related industries because if they are the number of selected securities has to increase to get the diversification benefit.

Rational risk averse investors will demand a higher return for a higher level of risk thus the expected return of a risky security is a function of the risk-free rate of return plus a risk premium. CAPM allows investors to measure the risk premium and it provides a method to begin to understand the market's risk and expected return curve.

CAPM focuses on systematic risk rather than total risk because unsystematic risk can be reduced or eliminated by diversification, so an investor is actually only rewarded with higher returns by assuming systematic risk.

Financial Theory and Financial Risk Management

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Beta is the accepted measure of systematic risk as it measures the return of a security against the market itself, which has a beta of 1. The S&P 500 index is often used as a proxy for the market, and any security with a beta higher than 1 will rise or fall at a greater rate than the market, whereas a beta less than 1 has a lower level of systematic risk and thus is less price sensitive to market swings.

This leads to what is often referred to as the security market line, which shows CAPM's risk and expected return relationship, with the focus being on systematic risk as measured by beta.

$$R_s = R_f + \beta_s(R_m - R_f)$$

R_s = the securities expected return

R_f = the risk free rate

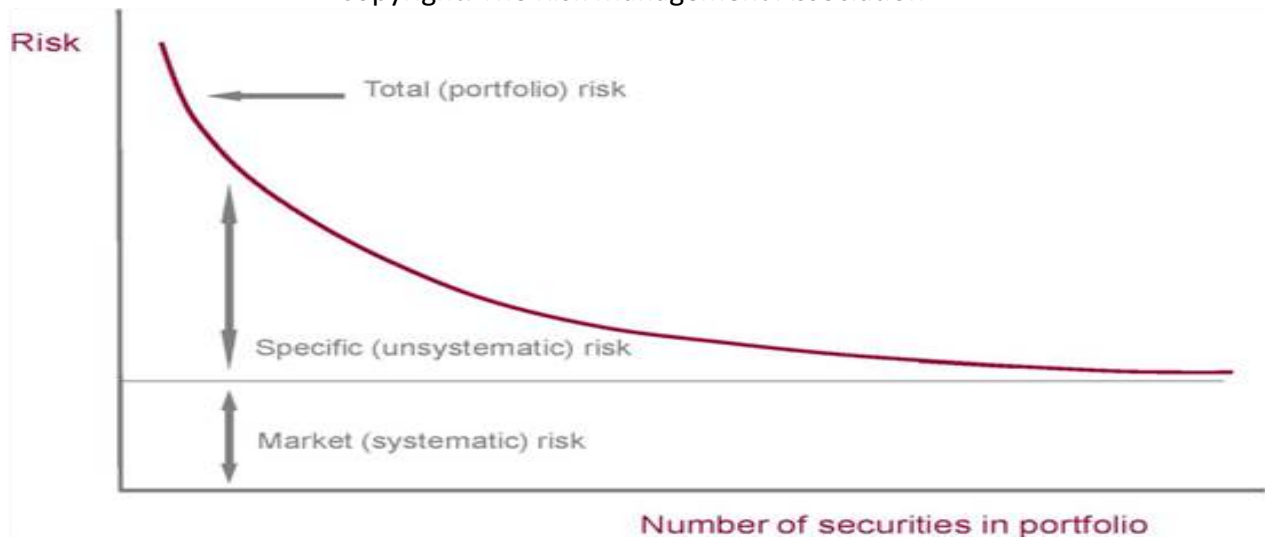
R_m = the expected return of the market itself

β_s = the beta of the security

(R_m - R_f) is the risk premium

The equation shows the relationship between risk and return, with the expected return being a function of the risk-free rate plus a risk premium, and in CAPM the risk premium is a function of beta times the market risk premium. Unsystematic risk is assumed to have been diversified away. Common drivers of systematic and unsystematic risk are in the following table and the image shows the diversification effect on unsystematic risk by adding additional securities.

Common Drivers of Systematic Risk	Common Drivers of Unsystematic Risk
• Monetary Policy • Fiscal Policy • Inflation • Economic Growth	• Management • Corporate Culture • Location • Industry



Like any model, CAPM is not perfect and it is built from limiting assumptions. But it also provides a useful way to measure what return an investor requires for a relative level of risk and CAPM can also be used in corporate finance, which defines the cost of equity as the expected return on a company's stock. If the company does not expect to meet its cost of equity (which can also be considered a hurdle rate) then it should return those funds to shareholders who can get this return from other securities.

Efficient Frontier

The efficient frontier is related to CAPM in that the efficient frontier represents the optimal set of portfolios that provide the highest expected return for a given level of risk. These portfolios are built from securities available in the marketplace, and the efficient frontier evaluates portfolios by two factors: return and risk, where the return is generally the Compound Annual Growth Rate of the security and its risk is the Standard Deviation. The efficient frontier was introduced by Nobel Laureate Harry Markowitz in 1952.

Unlike the security market line, which is a linear line, the efficient frontier is curved because it reflects the impact diversification has on the portfolio's risk / return profile, and it also shows there is a diminishing marginal return on risk. Adding additional risk to a portfolio does not necessarily mean an investor gets an equivalent return.

Markowitz's original theory included the construction of an optimal portfolio that has the perfect balance between risk and return, and it is constructed by balancing securities with the highest potential returns at an acceptable level of risk, or securities with the lowest potential return at the lowest risk. The data points on the plot of risk and return where the optimal portfolios lie are called the efficient frontier.

Capital Market Line

The capital market line represents those portfolios that optimally reflect risk and return. In theory as part of CAPM investors will select a position on the capital market line because it maximizes return at a given level of risk. Like the security market line, the capital market line is also linear.

The capital market line is different from the efficient frontier because it includes risk-free investments. The point of tangency between the capital market line and the efficient frontier is considered the most efficient portfolio. CAPM is the linear line that connects the risk-free rate with the tangency point on the efficient frontier, and this tangency point is also called the market portfolio.

$$R_p = r_f + \frac{R_t - r_f}{\sigma_T} \sigma_p$$

R_p = portfolio return

r_f = risk free rate

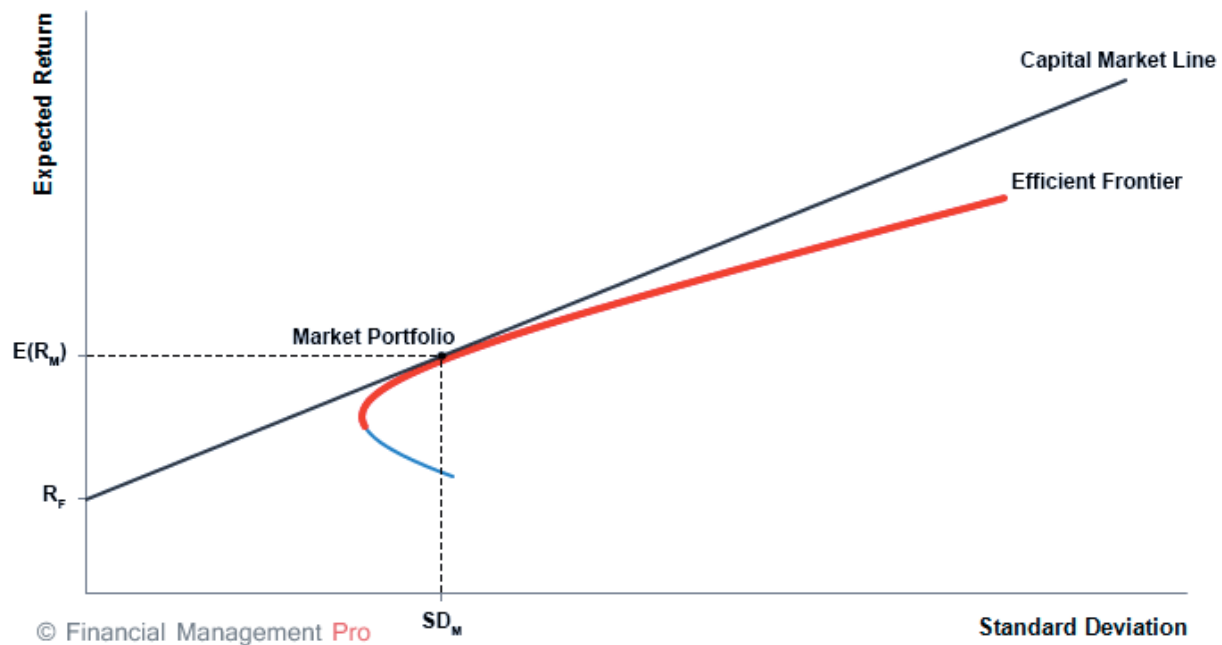
R_t = market return

σ_T = standard deviation of market returns

σ_p = standard deviation of portfolio returns

The capital market line shows the rates of return for a specific portfolio. As a comparison, the security market line is the market's risk / return at a point in time, and it shows the expected returns of individual investments.

Fairly priced securities will be on both lines, but securities above both lines have returns that are too high relative to risk and are considered underpriced. Securities below both lines have returns that are low relative to risk and are overpriced. The following image shows the capital market line, the efficient frontier, and the point of tangency that represent the market portfolio.



Conclusion

While CAPM, the security market line, efficient frontier, and the capital market line have limitations, they do succeed in allowing investors, traders and risk managers to evaluate risk and return and provide a means of comparison, as well as a benchmark and hurdle rates for analytical purposes.

Like any model they have limiting assumptions that need to be understood to ensure that the application of financial theory can be used to improve decision-making. Model risk management is becoming increasingly important as more models continue to be developed to support financial theory, trading and investment activity, and risk management. Models, however, should be used to support and improve decision-making and are tools available to investors, traders, and risk managers but we need to understand the models and how they are applicable in order to use them for the maximum benefit.